Utsav Sharma

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EDUCATION

Indian Institute of Technology Roorkee 2023 – 2027
Integrated B.S.–M.S., Mathematics and Computing CGPA: 8.05 / 10
Class XII (CBSE): 94.0% · Class X (CBSE): 95.4%

TECHNICAL SKILLS

Languages: Go, Rust, TypeScript, Python, SQL, Noir, Solidity
Backend & Distributed Systems: REST APIs, OAuth 2.0, Webhooks, Temporal workflows, Event-driven architecture, Idempotency
Databases: PostgreSQL, pgvector, Neo4j, Firestore, Schema design & migrations, Indexing, Write-ahead logging
Cryptography: Zero-knowledge proofs, zkVMs (Jolt, SP1, RISC Zero), R1CS, Poseidon2, Polynomial commitments (WHIR)
Tooling: Docker, GCP, GitHub Actions, Linux, Git, Claude Code, Codex

EXPERIENCE

Software Engineer | Trinity (AI project manager for software teams) Jul 2026 – Oct 2026
- Built the shared integration layer in **Go** behind GitHub, Linear, Slack, Granola and Notion, and the Notion integration itself.
- Built a common **meeting-notes layer** in PostgreSQL for every note-taking tool, cut over in three releases, and the Meetings backend on it: calendar feed, post-meeting task suggestions and briefs.
- Shipped task and milestone **completion forecasts** from a survival model refit nightly per project; beat the default in a staging backtest (Brier 0.114 vs 0.248).
- Shipped the **pre-standup brief**, a personal Slack summary on Temporal that reads and messages only its owner.
- Authored 157 pull requests (93 merged), directing **Claude Code and Codex** agents in parallel.

Protocol Engineer | Atheon May 2026 – Jun 2026
- Built a prototype **Jolt zkVM** prover on the hash-based **WHIR** commitment over the Goldilocks field that proves real RISC-V programs, and the design for moving Jolt off Dory.

ZK Cryptography Engineer | Atheon, on World Foundation's ProveKit Jan 2026 – Apr 2026
- Built zero-knowledge **passport age verification** in Noir as a chain of 4 to 5 circuits linked by Poseidon2 commitments, roughly halving peak proving memory (1.84 GB to 982 MB).
- Fixed two **passport-forgery vulnerabilities** from a security audit (unbound RSA exponent, smuggled signing key); upgraded Noir from beta.11 to beta.19.

Ethereum Protocol Fellow | Ethereum Foundation, Cohort 6 Jun 2025 – Nov 2025
- Built a **Rust** benchmarking framework for Ethereum consensus state transitions across 5 zkVMs (run time, cycles, proving cost), including the Jolt, Pico and Zisk backends.
- Prototyped a block-proving service in the **Ream** consensus client that gossips proofs to peers.

Software Engineer Intern | Enclave Jul 2025 – Nov 2025
- Built **MagicSpend++**'s Solana **Anchor** vault with signed, nonce-checked borrows, its Circle Gateway borrow path, one-transaction **EIP-7702** borrow-and-spend and USDC or SOL settlement, on 4 EVM mainnets and Solana devnet.
- Integrated it with **Privy and Turnkey** wallets and with **Fireplace**, a Polymarket trading app, for bets funded from Solana.

PROJECTS

SQL Database Engine | Rust · github.com/x-senpai-x/sqlDbEngine May 2026 – Jun 2026
- Wrote a SQL database from scratch with a parser, query planner, slotted pages, **B+ tree** index and LRU-K buffer pool.
- Added ARIES-style crash recovery on a **write-ahead log**; serves about 220K point reads per second on 10K rows.

Deimos | Rust, Flutter, Next.js · github.com/BlocSoc-iitr/Deimos Sep 2025 – May 2026
- Built the Rust proving core and its **Android and iOS** bindings for benchmarking zero-knowledge proofs on phones, comparing 6 hash circuits across Circom and Noir.

ACHIEVEMENTS

- **Global Rank 91**, Codeforces Round 950; **Expert** (max rating 1673), handle `frustratedaf`.
- **Winner**, ETHOnline 2024 and Agentic Ethereum 2025 (ETHGlobal); **Finalist**, Based India Buildathon 2024.
- Recipient, **INSPIRE Scholarship for Higher Education**, Government of India, for academic performance.

POSITIONS OF RESPONSIBILITY

- **Blockchain Society IITR** (Feb 2024 – Present): organised the StackTooDeep and Srishti developer events; ran technical sessions.
- **Finance Club IITR** (Feb 2024 – Present): organised Finvista and Quantathon; ran workshops on probability and algorithmic trading.